

Contractor Scraper Agent

Technical Specification & Build Brief

Version 2.0 · Florida Drywall, GC, Painter, Handyman Lead Generation

1. Project Overview

Build a data collection agent that produces a clean, deduplicated, tiered list of drywall, general contracting, painting, and handyman businesses operating in six Florida metro areas. Output should be ready for outbound sales outreach, with each row tagged by specialty tier and license status.

1.1 Target Geography

Florida metros (and ZIP codes within ~15 miles of city center):

- Tampa
- Orlando
- Daytona Beach
- Melbourne
- Jacksonville
- Gainesville

1.2 Target Business Types — Tiered

All four tiers below are IN SCOPE. Tier number drives sales priority, not inclusion/exclusion. Both licensed and unlicensed operators are included.

Tier	Specialty	Signals (Google category, business name, services list)
Tier 1	Drywall & texturing specialists	drywall, sheetrock, gypsum, plasterer, taping, mudding, popcorn ceiling, knockdown, orange peel, skim coat, level 5 finish, drywall texturing, drywall repair, drywall finishing
Tier 1	General contractors	Google type general_contractor AND any drywall keyword present in name, services, or business description
Tier 2	Painters & finish carpentry	Google type painter; OR keywords: painting contractor, interior painter, residential painter, drywall and paint, paint and patch
Tier 2	Remodelers & home repair	Keywords: remodeling, renovation, home improvement, interior remodel, residential renovation (and not in exclusion list)
Tier 3	Handymen / general repair	Keywords: handyman, handyperson, home repair, property maintenance, small repairs

Tier	Specialty	Signals (Google category, business name, services list)
EXCLUDE	Hard exclusions	HVAC, air conditioning, AC repair, heating, asphalt, paving, concrete pouring, concrete contractor, landscaping, lawn care, tree service, pool, pool cleaning, pest control, septic, well drilling, fence only, roofing only, plumbing only, electrical only, locksmith, garage door only, window only, flooring only, cleaning service, junk removal

Exclusion rule: A business is excluded ONLY if it matches an exclusion keyword AND has zero Tier 1/2/3 keywords. A roofer who also does drywall repair stays in. A pure roofer is out.

1.3 Licensed vs. Unlicensed

Both are in scope. Florida technically requires a Gypsum Drywall Contractor license, but a significant share of the working drywall and texturing market operates without one. We cross-reference the DBPR database to TAG each business with license status, but we do not filter based on it. Sales will segment from there.

The license_status field uses these values:

- licensed_active — found in DBPR, Current/Active status
- licensed_inactive — found in DBPR, but Inactive/Delinquent/Null and Void
- unlicensed — has Google Business presence but no DBPR match
- unknown — DBPR match uncertain (e.g. fuzzy name match below confidence threshold)

1.4 Final Deliverable

A single deduplicated master CSV (and JSON mirror) where each row represents one contractor business, tagged by city, ZIP, tier, license status, and specialty. Per-city files as well. Schema is defined in Section 5.

2. Data Sources & Order of Operations

Strategy: Google Maps is the primary DISCOVERY source. DBPR is an ENRICHMENT step that tags license status. BBB is supplementary.

2.1 Source A — Google Maps (PRIMARY)

Google Maps is the only source that captures both licensed and unlicensed operators. Most one-truck drywallers, painters, and handymen have a Google Business profile but no state license. Use one of the two approaches below — do not build a raw scraper from scratch.

Recommended: Outscraper Google Maps Scraper (or Apify equivalent)

Why this over the official Places API:

- No 60-result cap per query (the API caps; Outscraper does not)
- Returns emails scraped from the linked business website in the same pass

- Lets you filter by Google's native category subtypes ("drywall contractor", "painter", "general contractor") with exact-match logic
- Cheaper: roughly \$30 per 10,000 leads vs. ~\$32 per 1,000 official-API requests
- Returns additional fields: social profiles, owner-written descriptions, services list, "from the business" sections

Native Google category subtypes to target (Outscraper exposes these as filterable):

```
drywall_contractor, general_contractor, painter, painting, plasterer,
ceiling_contractor, wallpaper_contractor, finish_carpentry_contractor, handyman,
remodeler, home_improvement_store
```

Query plan (per city):

Run each of the following search phrases against each ZIP code in the city. Outscraper will multiply queries × locations automatically.

```
drywall_contractor drywall repair drywall texturing sheetrock contractor plasterer popcorn
ceiling general contractor painting contractor painter remodeling contractor home
renovation handyman home repair
```

Outscraper advanced filters to apply at scrape time:

- subtype CONTAINS one of: drywall, general contractor, painter, plasterer, ceiling, handyman, remodel, home improvement
- subtype CONTAINS NONE of: hvac, air conditioning, asphalt, paving, concrete, landscap, pool, pest, roofing only, plumbing only, electrical, tree, septic, locksmith
- business_status = OPERATIONAL
- phone IS NOT BLANK (optional — drop if you want to keep listings without phone for enrichment)

Cost estimate: Assuming roughly 3,000–5,000 unique businesses across all six metros after dedupe, expect to scrape 15,000–25,000 raw rows. At Outscraper rates, total ~\$45–\$75 for the first full run.

Alternative: Google Places API (only if Outscraper is rejected)

If your team has compliance reasons to stay on the official API, plan for higher cost, more code complexity (pagination over Text Search + Nearby Search + multiple radius rings to work around the 60-result cap), and no email enrichment in the same pass.

2.2 Source B — Florida DBPR (ENRICHMENT, not gate)

The Florida Department of Business & Professional Regulation maintains a public license database. Pull the full list of license-holders in our target categories, then match against the Google results.

Public license search URL:

```
https://www.myfloridalicense.com/wl11.asp?mode=2&search=Name
```

License categories to pull (broad — match all of these):

- Gypsum Drywall Contractor (Certified and Registered)
- Certified General Contractor (CGC)
- Certified Building Contractor (CBC)
- Certified Residential Contractor (CRC)

- Registered General/Building/Residential Contractor (RG/RB/RR)
- Painting Contractor (where listed as a specialty)

Matching logic:

1. Try exact match on normalized business name + city
2. Try exact match on normalized phone number
3. Try fuzzy match on business name (rapidfuzz token_set_ratio $\geq$ 88) within same ZIP
4. If multiple licenses match one Google listing, attach all license_numbers as a list
5. Set license_status based on match outcome (see Section 1.3)

Implementation: DBPR uses ASP postbacks. Use Playwright with 2–3 second delays. Cache HTML responses. Pulling the full license set for all relevant categories statewide is feasible in one overnight run — Florida has roughly 130,000 contractor licenses total across all categories.

2.3 Source C — Better Business Bureau (Supplementary)

BBB has decent coverage of established contractors but misses most solo operators and unlicensed handymen. Use it only to enrich the matched list with:

- BBB rating (A+, A, etc.) — useful sales signal for tier-1 outbound
- BBB accreditation status
- Years in business
- Detected closures/out-of-business flags

Implementation: Use an existing Apify or Outscraper BBB actor rather than building from scratch — BBB actively blocks raw scrapers. Cost is negligible (~\$5–\$15 for the enrichment pass).

Search URL pattern:

```
https://www.bbb.org/search?find_country=USA&find_loc={city}%2C+FL&find_text=drywall
```

3. Architecture & Pipeline

Sequential pipeline. Each stage writes to its own JSONL file so any stage can be re-run independently.

3.1 Pipeline Stages

6. Stage 1 — Google Discovery. For each (city, ZIP, query phrase) tuple, call Outscraper. Apply subtype include/exclude filters at scrape time. Save raw results to stage1_google.jsonl.
7. Stage 2 — DBPR Pull. For each of the six metros, query DBPR for every relevant license category. Save to stage2_dbpr.jsonl.
8. Stage 3 — Tier Assignment. For each Google row, run the keyword classifier (see 3.3) to assign Tier 1/2/3 or EXCLUDE. Drop EXCLUDE rows.
9. Stage 4 — License Match. Cross-reference each remaining row against DBPR. Set license_status and attach license_numbers.
10. Stage 5 — Dedupe. Merge duplicates by phone, then website domain, then fuzzy name + ZIP. Keep most complete record.
11. Stage 6 — BBB Enrichment (optional). Look up BBB rating/accreditation for matched rows.

12. Stage 7 — Contact Enrichment (optional). Push rows missing email through Apollo.io or Hunter.io.
13. Stage 8 — Export. Write master CSV/JSON plus per-city files.

3.2 Recommended Stack

Component	Recommendation
Language	Python 3.11+
Google scraping	Outscraper Python SDK (preferred) or Apify Python client
DBPR scraping	Playwright (headless Chromium) with 2–3s throttle
BBB scraping	Apify or Outscraper BBB actor (do not roll your own)
Data store	JSONL per stage + SQLite for the final merged dataset
Dedupe / matching	rapidfuzz for fuzzy name matching, phonenumbers for phone normalization, usaddress for address normalization
Enrichment	Apollo.io API or Hunter.io for emails; optional People Data Labs
Scheduling	Cron or APScheduler for monthly re-runs
Output	Master CSV + JSON; per-city CSV + JSON (6 each)
Logging	loguru with per-stage row counts and reject reasons

3.3 Tier Classifier Logic

Apply this pseudocode to every Google result. Order matters — exclude first, then check tiers from most specific (1) to least (3).

```
def classify(row):
    text = (row.name + " " + row.subtypes + " " + row.services + " " + row.description).lower()
    DRYWALL_KW = ["drywall", "sheetrock", "gypsum", "plaster", "taping", "mudding", "popcorn ceiling", "knockdown", "orange peel", "skim coat", "drywall texturing", "drywall repair", "level 5"]
    GC_KW = ["general contractor", "general contracting", "construction services"]
    PAINT_KW = ["painter", "painting", "painting contractor", "residential painter"]
    REMODEL_KW = ["remodel", "renovation", "home improvement", "home renovation"]
    HANDYMAN_KW = ["handyman", "handyperson", "home repair", "property maintenance"]
    EXCLUDE_KW = ["hvac", "air conditioning", "ac repair", "heating", "asphalt", "paving", "concrete pouring", "concrete contractor", "landscap", "lawn care", "tree service", "pool", "pest control", "septic", "well drilling", "locksmith", "garage door", "junk removal", "cleaning service"]
    # Solo-trade exclusions (only if NOTHING in scope is also present)
    SOLO_EXCLUDE = ["roofing", "plumbing", "electrical", "flooring", "tile", "window", "fence"]
    has_target = any(k in text for k in DRYWALL_KW + GC_KW + PAINT_KW + REMODEL_KW + HANDYMAN_KW)
    has_hard_exclude = any(k in text for k in EXCLUDE_KW)
```

```
has_solo_exclude = any(k in text for k in SOLO_EXCLUDE)    if has_hard_exclude and
not has_target:      return "EXCLUDE"    if has_solo_exclude and not has_target:
return "EXCLUDE"    if any(k in text for k in DRYWALL_KW):      return
"TIER_1_DRYWALL"    if any(k in text for k in GC_KW) and any(k in text for k in
DRYWALL_KW + PAINT_KW + REMODEL_KW):      return "TIER_1_GC_WITH_SCOPE"    if any(k
in text for k in GC_KW):      return "TIER_2_GC_GENERIC"if any(k in text for k in
PAINT_KW):      return "TIER_2_PAINTER"    if any(k in text for k in REMODEL_KW):
return "TIER_2_REMODELER"    if any(k in text for k in HANDYMAN_KW):
return "TIER_3_HANDYMAN"      return "EXCLUDE"
```

3.4 Deduplication

In priority order — first match wins:

- 14. Exact match on normalized phone number (strip all non-digits, drop leading 1)
- 15. Exact match on normalized website domain (strip www, trailing slash, query strings)
- 16. Fuzzy match on business name (rapidfuzz token_set_ratio ≥ 90) AND same ZIP
- 17. Exact match on normalized full address

When merging, keep the most complete record. Store all source place_ids and license_numbers as lists so we can trace back.

4. Data Enrichment (Phase 2)

Once the master list is clean, enrich rows missing email, owner name, or LinkedIn. Cap spend at roughly \$0.30 per enriched record.

Provider	Best for	Cost (approx)	Notes
Apollo.io	Owner emails, LinkedIn URLs	\$49–\$99/mo starter	Best SMB construction coverage
Hunter.io	Email from domain	\$49/mo, 500 searches	Cheapest if email is all you need
People Data Labs	Person + company enrichment	~\$0.05–\$0.20 per match	Pay-as-you-go API; strong fallback
Outscraper Email Finder	Email from website (bundled)	Included in Google scrape	Use first — many rows resolved free

Recommended order: Outscraper's bundled email finder during the main scrape resolves a large share of rows for free. Then Hunter.io for any business with a website but no email. Then Apollo.io for owner-name and LinkedIn. People Data Labs as last-resort fallback for unresolved rows.

5. Output Schema

Deliver as CSV (primary) and JSON (mirror). All columns below are required (use null if unknown).

Field	Type	Description
business_name	string	Cleaned legal or DBA name
city	string	Metro the row was discovered under
zip_code	string	5-digit ZIP of physical address
address	string	Full street address
tier	string	TIER_1_DRYWALL / TIER_1_GC_WITH_SCOPE / TIER_2_GC_GENERIC / TIER_2_PAINTER / TIER_2_REMODELER / TIER_3_HANDYMAN
specialty_keywords	list	Matched keywords used to classify (transparency for sales)
google_categories	list	Google subtypes from the business profile
services_listed	list	Services from the Google Business profile (when available)
phone	string	E.164 format, e.g. +18135551234
email	string	Best email found across sources
website	string	Normalized URL (https, no trailing slash)
owner_name	string	From DBPR licensee or Apollo enrichment
license_status	string	licensed_active / licensed_inactive / unlicensed / unknown
license_numbers	list	All DBPR license numbers matched to this business (may be empty)
license_categories	list	e.g. ["Gypsum Drywall", "CGC"]
google_rating	float	Average Google rating (0–5)
google_review_count	int	Number of Google reviews
bbb_rating	string	BBB grade (A+, A, B, etc.) if listed
bbb_accredited	bool	Whether the business is BBB-accredited
years_in_business	int	From DBPR original-issue date, BBB, or Google when available
social_profiles	dict	Facebook, Instagram, LinkedIn URLs from website footer scrape

Field	Type	Description
sources	list	["google", "dbpr", "bbb", "apollo"] — which contributed to this row
place_ids	list	All Google place_ids that merged into this row
scraped_at	datetime	UTC timestamp of last scrape

6. Legal & Compliance Notes

- **DBPR data:** Public record under Florida's Sunshine Law (Chapter 119, F.S.). Safe to scrape and store.
- **Google Maps data via Outscraper/Apify:** Federal courts have held (*hiQ v. LinkedIn*, *Meta v. Bright Data 2024*) that scraping publicly visible data is not a CFAA violation. It does technically breach Google's Terms of Service, which is a private contract, not federal law. The legal risk is account-level (Google can block IPs), not criminal. Have counsel review before large-scale resale of the dataset.
- **BBB data:** Public business profiles. Use an existing actor to avoid IP blocks.
- **Email outreach:** CAN-SPAM applies — every outbound email needs clear sender identity, a physical mailing address, and an opt-out mechanism.
- **Do NOT collect:** Personal SSNs, license application financials, anything beyond what is publicly displayed on a business listing.
- **Developer agreement:** Have the developer sign an NDA + work-for-hire clause before sharing this spec. The source code, the API keys, and the resulting dataset must all be yours.

7. Deliverables & Acceptance Criteria

7.1 Deliverables

- Git repository with Python source, README, requirements.txt
- config/cities.yaml — list of six cities with their ZIP codes (you provide; dev wires up)
- config/keywords.yaml — the tier classifier keyword lists (editable by you without code changes)
- Single entry point: `python -m agent.run --full` or per-stage flags
- Master CSV: `contractors_florida_master.csv`
- Per-city CSV: `contractors_{city}.csv` (six files)
- Per-city JSON mirror of each CSV
- Run log showing rows in/out per stage and reject reasons
- README under 3 pages covering: setup, env vars, how to re-run, how to edit keywords

7.2 Acceptance Criteria

- At least 2,000 unique contractor businesses in the master file across the six metros

- At least 30% of rows in Tier 1 (drywall or GC with drywall scope)
- $\geq 90\%$ of rows have a working phone number
- $\geq 60\%$ of rows have a website OR an email
- Every row has tier and license_status populated
- 0 rows match a hard exclusion keyword (HVAC, asphalt, concrete-pouring, etc.)
- 0 duplicate rows by normalized phone or normalized domain
- Full pipeline re-run completes on a fresh machine in under 6 hours

8. Phasing & Timeline

Week 1: Spec sign-off, NDA, API accounts created (Outscraper, Apollo, Hunter). You own them, you share scoped credentials with the dev.

Week 2: Milestone 1 — Google scrape + tier classifier running for Tampa. You review output for accuracy.

Week 3: Milestone 2 — All six cities, DBPR cross-match, dedupe. Review master file.

Week 4: Milestone 3 — BBB + Apollo enrichment, documentation, final delivery.

Month 2+: Monthly re-runs to catch new businesses and license changes. Optional retainer.

9. Open Questions for You

Decide these before kickoff so the scope is locked:

18. Do you want explicit ZIP lists per metro, or let Outscraper handle radius-based search from city center? (Recommend: explicit ZIPs — more controlled output.)
19. Is email enrichment in scope for v1, or Phase 2? (Recommend: include Outscraper's free bundled email finder in v1; defer Apollo/Hunter to Phase 2.)
20. Should Tier 3 (handymen) be tagged-but-deprioritized in the output, or actually excluded from the export? (Recommend: include with tier label so sales can decide.)
21. Who owns the API keys for Outscraper, Apollo, Hunter? (Strong recommend: YOU. Dev uses your keys, not the other way around.)
22. How often should the pipeline re-run after delivery — weekly, monthly, on-demand?
23. Do you need a simple web UI to filter/browse the master list, or is CSV/JSON enough?